Umar Shahzad

Electrical Engineer | Data Center Engineering Operations
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SUMMARY

Electrical engineer in Data Center Engineering Operations (DCEO) at AWS Milan, working safety-first on live critical infrastructure: generators, UPS, MV/LV switchgear, circuit breakers, transformers and PDUs. Raised 3 safety findings, supported 8 rack Power Ups/Preps and am building a redundancy simulator for 3 sites. B.E. Electrical Power Engineering (ranked 3rd of 120), MSc Energy Engineering. Relocating to Frankfurt; available January 2027.

WORK EXPERIENCE

Data Center Engineering Operations (DCEO) Intern Sep 2026 – Nov 2026
Amazon Web Services (AWS) *Milan, Italy*

- Raised 3 safety findings as a new joiner: stopped a vendor working near energised equipment without eye protection until PPE was worn, reported a jammed hot-aisle door blocking emergency egress, and corrected missing gloves in the data hall; all remediated, recognised by the site team.
- Supported engineers on 3 rack Power Ups and 5 Power Preps: verified pre-energisation checklists, matched circuits and confirmed space clearance before rack placement; all 8 completed as planned with no deviations.
- Building a redundancy simulator for 3 Milan sites that models each site's redundancy configuration; users fail any UPS, generator or switchgear with one click and see load redistribution. Turns tabular colocation reports into per-line loading (%) against capacity, standardising how the team checks load and trains on failure scenarios; AI with RAG keeps logic tied to site data. Delivery: early Nov 2026.
- Completed AWS EOT and Global Safety training covering generators, UPS, MV/LV switchgear, PDUs and data center electrical distribution.

Thesis Researcher (Erasmus+ Exchange) Apr 2026 – Aug 2026
Schering Institute (High Voltage & Asset Management), Leibniz University Hannover *Hannover, Germany*

- Thesis: *Vibration and Noise Behaviour of Transformer Cores under Electromagnetic Excitation*. Ran experiments on transformer core samples across 12+ clamping-force configurations up to 2000 Hz; validated COMSOL Multiphysics FEM models against measured vibration spectra with less than 5% deviation, supporting HV asset condition monitoring.

Proposal Engineer – EPC (Energy Infrastructure) May 2024 – Oct 2024
Barqtron Engineering Solutions *Karachi, Pakistan*

- Managed proposal development for three live 132 kV transmission and grid-station projects in competitive government tenders; compliance reports secured bid qualification in all three tender rounds.
- Assessed 220/132 kV grid stations and transmission routes on site to define scope; reviewed electrical designs and equipment for compliance.
- Coordinated USD 5M procurement of power transformers, HV switchgear and cables across 4 suppliers; produced AutoCAD BOQs and cable schedules and Primavera P6 schedules for civil, installation and commissioning works.

Automation Intern – PLC, SCADA & IIoT Aug 2023 – Sep 2023
Automation Play *Remote*

- Programmed PLCs and configured SCADA interfaces (Factory I/O) across 3 control and automation projects.

Electrical Engineering Intern – 500/220 kV Grid Station Jun 2023
National Transmission & Dispatch Company (NTDC) *Karachi, Pakistan*

- Supported preventive maintenance of 500/220 kV power transformers, circuit breakers, CTs/PTs and HV switchgear; studied distance, differential and Buchholz protection schemes in live HV operations.

EDUCATION

M.Sc. Energy Engineering Sep 2024 – Dec 2026 (expected)
University of Padua, Italy | Erasmus+ Exchange: Leibniz University Hannover *Padova, Italy*
Coursework: Electric Power Systems; Technologies for HVAC and HVDC Transmission Systems; Smart Grids; Measurements and Instrumentation; Heat Transfer and Thermofluid Dynamics.

B.E. Electrical Engineering (Power) Sep 2019 – Jul 2023
Bahria University | CGPA 3.86/4.0 | Magna Cum Laude, ranked 3rd of 120 *Karachi, Pakistan*
Coursework: Power System Protection; Power Distribution & Utilization; Power Generation; Electrical Power Transmission; Electrical Machines; Power Electronics; Linear Control Systems.

SKILLS

Critical Facility Equipment: UPS, diesel generators, MV/LV switchgear, circuit breakers, power transformers, PDUs, electrical distribution, redundancy configurations, protection schemes
Operations & Safety: Safety inspections and hazard reporting, PPE enforcement, preventive maintenance (PPM), pre-energisation checks, rack Power Up/Power Prep, procedure compliance, contractor and vendor coordination, Primavera P6 scheduling
Tools: AutoCAD, ETAP, PowerFactory, MATLAB/Simulink, COMSOL, Python, PLC/SCADA, AI/RAG app development

CERTIFICATIONS & TRAINING

AWS EOT Training • AWS Global Safety Training • AI Navigator for DC Certification • OSHA Workplace Safety Training • Jenbacher 420 Gas Engine O&M • ETAP Certification • AutoCAD Certification

LANGUAGES

English – C1 • Italian – A2 • German – A1